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7/6/25

Applied Cyber Threat Analysis – Malware Analysis Lab

- **This lab was conducted within the Virginia Cyber Range Environment.**
- **Report based off assignment assigned by George Mason University - Department of Information Sciences and Technology College of Engineering and Computing**
- **Cheche Agada. (2024). Malware Analysis Lab IT-462: Applied Cyber Threat Analysis, George Mason University**

Introduction
In this lab, we will focus on conducting a static analysis of some unknown and suspicious files. In Task 1 you will be given an executable file to analyze. Analyze the file as best you can and record your results in the space provided.
In Task 2, you will use an open-source tool (virus total) to investigate file hashes created from suspicious files**.**

Objective
Conduct static analysis of suspicious files.

Example tools that can be used to analyze selected suspicious files

- HashCalc
- CFF_Explorer
- Pestudio
- PEview
- PEiD
- bintext

Task 1

Using the file you have selected, conduct a static analysis and record your results in the template provided below:

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STATIC ANALYSIS

Name of Investigator:	Troy Fracyon
Date:	7/6/2025
Source:	Malware Analysis – Unknown Code.zip
File Number:	13
File Location:	C:\Users\student\Desktop\13.exe
File Timestamps:	7/6/2025 8:06pm
Original File Name:	PuTTy
File Size (bytes):	1.23 MB
MD5 Hash	14080A3E4E877BE235F06509B2A4B6A9
SHA 1 Hash	868866BD51F1AC744991C08EDA6446222A0CCDAE
File Type	Portable Executable 64
OS Environment	Windows
Version (Language):	English
Dependencies:	GDI32.dll, IMM32.dll, ole32.dll, USER32.dll, KERNEL32.dll, SHELL32.dll, COMDLG32.dll, ADVAPI32.dll
Programming Language:	Microsoft Visual C++ 8.0 (DLL)
Compiler Timestamp:	Friday 28 October 2022, 17.23.40/Fri Oct 28 17:21:05 2022 (UTC)
Date Modified:	Sunday 06 July 2025, 20.34.33
Packed (Y/N):	Y
Strings:	“RandSeedFile”, ”Software\SimonTatham\PuTTY” “Software\SimonTatham\PuTTY\SshHostKeys”
OSINT (VT, Google, etc.):	Summary: PuTTy is an open-source SSH and Telnet client for Windows systems.

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	Main webpage: https://putty.software/ Project homepage: https://www.chiark.greenend.org.uk/~sgtatham/putty/ Recent Security Updates: Version 0.84 (May 2026) fixed: ● RSA key exchange double-free vulnerability. ● ECDSA verification crash issue. ● Telnet/Rlogin trust-signaling issue. Security History: CVE-2024-31497 - critical cryptographic vulnerability. Allows for attackers to recover a user's NIST P-521 private keys. Affected versions (.68-.80)

Task 2

LAB: Virus Total

www.virustotal.com

Instructions: Investigate the following files on virustotal and record your observations

File name	File hash (MD5)	Observations
Index.exe	113A1A35C7EB4F017DD13FC4D0613837	62/71 flagged, 332.48 KB, windows environment, UPX anti-analysis, keystroke capturing, drops files in directory
Salaries.exe	F59857d881e848f8b9442fb1ae3066b5	62/71 flagged, 808.82 KB, windows executable, 6 marks on domain vboxsvr.ovh.net, command line options/arguments, must delete registry key, capable to modify access privileges, XOR encode data, logs keystrokes

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tripItenerary.exe	EF961702407831EBF6AE723E3B1A8BC2	61/72 flagged, 332.30 KB, windows executable, interacts with registry to hide information, persistent payload through system process modification, UPX obfuscation
Grades.exe	43D2D041D9F52137A6F3D4698B374984	59/72 flagged, trojen, worm, downloader, 808.78 KB, windows executable, command line arguments, delete registry key, XOR and blowfish obfuscation, keystroke capture
Pleasantly surprised.exe	5A5B8B5F973ECE3D4491F9F7C27068BC	54/72 flagged, detects debug, self-deletes, 552 kb, windows executable, C++, created 2015-09-02, 4/5 marks on URL/domain meitanjiaoyiwang[.]com, code injection privilege escalation, can collect keystrokes and obfuscate self, file manipulation.
Evil.exe	E696B38AC71B23F50EE68DA06A004AF3	60/72 flagged, 675.13 KB, trojan, dropper, windows executible, c++, created 2013-08-22, many flagged URLs and domains associated including, wike.aikaba,com and a.sinkhole.yourtrap,com, code injection privilege escalation